OSTAD | DEVOPS | GIT ASSIGNMENT

Nanu's Recipe Book

A story-driven journey through Git

Total Marks	Submit	Covers
100	GitHub repo link + my-story.md	Setup, Branching, Remote, Stash, History

The Story

Your Nanu (grandmother) has an old handwritten recipe book full of family recipes: bhapa pitha, Eid biryani, shemai and more. The pages are tearing, so you've decided to make it digital, with help from your cousin Mitu in Chattogram. There's only one rule: **no recipe can ever be lost.** That's why you'll use Git.

For each chapter, write the commands you ran and what happened in my-story.md, in your own words, like a story.

Chapter 1 The First Recipe (20 Marks)

Nanu says: "Winter is coming. Write the bhapa pitha recipe first."

Task A Set your name and email with `git config --global`, then create a folder nanur-khata and run `git init`.

Task B Create bhapa-pitha.md, check `git status`, and stage it with `git add`.

Task C Nanu corrects you: "One and a half cups of gur, not two!" Edit the file, run `git diff` and `git diff --staged`, and explain the difference.

Task D Commit with a meaningful message using `git commit -m`.

Task E You accidentally staged bazar-list.txt. Unstage it with `git reset [file]` without deleting it.

Chapter 2 The Secret Spice (10 Marks)

Nanu's secret spice blend must never reach GitHub.

Task A Create secret-masala.txt, a few .tmp files and a draft/ folder.

Task B Write a `.gitignore` that keeps them all out, and prove it with `git status`.

Chapter 3 The Eid Special (20 Marks)

You want to test the Eid recipes without touching the main book.

Task A Create a branch with `git branch eid-special` and switch to it with `git checkout`.

Task B Add biryani.md and shemai.md in two separate commits.

Task C Go back to main. Are the files there? Then run `git merge eid-special` and check `git log`.

Challenge Create a merge conflict on purpose and resolve it. Explain how.

Chapter 4 Mitu Joins In (20 Marks)

Mitu wants to write in the book too, all the way from Chattogram.

Task A Create a GitHub repo, connect it with `git remote add origin [url]`, and `git push`.

Task B Play Mitu: `git clone` the repo into another folder, add chittagong-mezban.md and push.

Task C Back in your folder, run `git fetch origin`, then `git merge origin/main`.

Task D Mitu pushes again. This time use `git pull`.

Question: What is the difference between fetch + merge and pull?

Chapter 5 Half-Cooked Halua and Time Travel (20 Marks)

Halfway through the halua recipe, Nanu calls: "You forgot cardamom in the biryani!"

Task A Save the halua work with `git stash`, fix and commit the biryani, then `git stash pop`.

Task B Organize recipes into folders with `git mv`, delete an old recipe with `git rm`, and check `git log --stat -M`.

Task C Pick a commit and run `git show [SHA]`.

Task D On a test branch, try `git reset --hard [older-commit]` and write down what you lost. **Never do this on main!**

Question: Why are rebase and reset --hard dangerous on a shared branch?

Submission (10 Marks)

Rule 1 At least 10 meaningful commits and 2 branches in the repo. Messages like "update" lose marks.

Rule 2 my-story.md with your story, commands, key screenshots and answers.

Rule 3 End with a screenshot of `git log --oneline --graph --all`.

> **Warning:** If secret-masala.txt is found in your repo, Chapter 2 gets zero. Nanu's secrets stay secret!

Marking Scheme

Chapter	Topic	Marks
1	The First Recipe	20
2	The Secret Spice	10
3	The Eid Special	20
4	Mitu Joins In	20
5	Half-Cooked Halua and Time Travel	20
—	Submission Quality	10
Total		**100**